MOLE CONCEPT

- Relative atomic mass = $\frac{\text{mass of 1 atom of an element}}{1\ \text{amu}}$
(Dimensionless)

1 amu (or 1u) = $\frac{1}{12}$ th of mass of 1 atom of C-12 isotope

= 1.66×10^{-24} g

- Atomic mass = Mass of 1 atom of an element in amu

e.g Sulphur 'S' , RAM = 32

Atomic mass = 32 amu

- Gram atomic mass = mass of 1 mol of atoms of an element

= 32amu × N_A

= $32 \times 1.66 \times 10^{-24} \times 6.023 \times 10^{23}$

= 32g/mol

- Definition of mol:
Amount of substance which contain as many atoms/molecules/ions/or any other entity which are equal to no. of atoms in 12g of C-12 isotope.

1 mol = 6.022×10^{23} = Avogadro's Number (N_A)

Calculation of number of mole :

(1.) When no. of particles are given :

$$\text{No. of moles} = \frac{\text{Given no. of particles}}{N_A}$$

(2) When mass of substance is given

$$\text{No. of moles of substance} = \frac{\text{given mass (g)}}{\text{Molar mass (g/mol)}}$$

(3) For gases, if volume of gas is given under STP or NTP condition :

$$\text{No. of moles of gas} = \frac{\text{Given volume of gas (L) in STP}}{\text{Molar volume of gas } \left(\frac{L}{mol}\right) \text{ in STP}}$$

STP or NTP :

Pressure = 1 bar

Temperature = 0°C or 273.15 K ; $K = {}^{\circ}C + 273$

→ Molar volume of gas at STP = 22.4 lit/mol

⇓ By old definition of STP

⇓ By new definition of STP (22.7 lit/mol)

Q → Which is correct value of molar volume of gas at STP ?

(A) 22.4 lit (B) 22.7 lit (C) 25 lit (D) 24.63 lit

- If gases is not under STP / NTP condition.

$PV = nRT$ (Ideal gas eqⁿ)

R = universal gas constant

$= 0.821$ lit atm mol^{-1} k^{-1}

$= 8.314$ J mol^{-1} k^{-1} (SI unit)

$PV = nRT$

atm, lit, kelvin

0.0821 lit atm mol^{-1} k^{-1} $\approx \frac{1}{12}$ lit atm mol^{-1} k^{-1}

$PV = nRT$

Pa, m^3, kelvin

8.314 J mol^{-1} k^{-1} $\approx \frac{25}{3}$ J mol^{-1} k^{-1}

1 atm $= 1.01325 \times 10^5$ Pa (or N/m^2)

$\approx 1.013 \times 10^5$ Pa

1 bar $= 10^5$ Pa

1 atm $= 1.013$ bar

1 atm $= 760$ mm of Hg $= 76$ cm of Hg $= 0.76$ m of Hg

$= 760$ torr

- $1 m^3 = 1000$ lit

1 lit $= 1000 cm^3$

$1 m^3 = 10^6 cm^3$ or 10^6 ml

Q. 90 gm glucose ($C_6H_{12}O_6$)

Find (i) No. of moles of glucose
(ii) No. of moles of C, H, O atoms
(iii) Mass of C in it
(iv) No. of oxygen atoms in it.

(i) $n_{glucose} = \frac{90}{180} = 0.5$ mol

(ii) $n_C = 6 \times 0.5 = 3$ mol
$n_H = 12 \times 0.5 = 6$ mol
$n_O = 6 \times 0.5 = 3$ mol

(iii) mass of carbon $= 3 \times 12 = 36$ g

(iv) $3_{N_A} = 3 \times 6.022 \times 10^{23}$

Q. What mass of oxygen gas is present in 2 lit vessel at 5 atm pressure and 300K temperature

$PV = nRT$

$5 \times 2 = \frac{m_{O_2}}{32} \times \frac{1}{22} \times 300$

$\therefore m_{O_2} = 12.8$ g

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→ Atomicity :-
No. of atoms in molecule.

He, Ne, Ar ⟶ Monoatomic
H_2, O_2, N_2, CO, NO ⟶ Diatomic
SO_2, N_2O, H_2O ⟶ Triatomic

Q Crystals of copper sulphate $CuSO_4 \cdot 5H_2O$ (50g)
Find mass of water of crystallisation in it.

Sol^n $CuSO_4 \cdot 5H_2O$ (50g)

$$n_{CuSO_4 \cdot 5H_2O} = \frac{50}{64+32+64+90} = \frac{50}{250} = 0.2 \text{ mol}$$

$$= 5 \times 0.2 = 1 \text{ mol of water}$$
$$= 18\text{g} \quad \text{Ans}$$

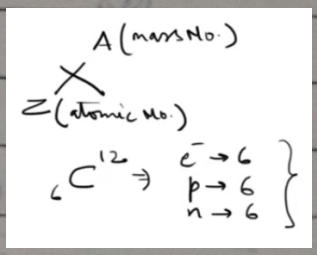
Q. $Na_2S_2O_3 \cdot 5H_2O$ (W g sample)

Mass of water of crystallisation

$$= \frac{W}{2\times23+2\times32+3\times16+5\times18} \times 5 \times 18 \text{ g} = \text{Ans}$$

Q. Methane, CH_4 (8g sample)
Find no. of moles of e^-, p, n in it

$$n_{CH_4} = \frac{8}{16} = 0.5 \text{ mol}$$



(CH_4)

$e^- \to 6 + 4\times1 = 10$

$p \to 6 + 4\times1 = 10$

$n \to 6 + 4\times0 = 6$

$$n_{e^-} = 10 \times 0.5 = 5 \text{ mol}$$
$$n_p = 10 \times 0.5 = 5 \text{ mol}$$
$$n_n = 6 \times 0.5 = 3 \text{ mol}$$

No of neutrons = $3N_A$ = $3 \times 6.022 \times 10^{23}$ neutrons
No of protons = $5N_A$
No of electrons = $5N_A$

Q. 10 mol of H_3PO_4 + 2 mol P_4 + 1 mol $H_4P_2O_7$

All phosphorus 'P' is taken out & converted into P_4. Find no. of moles of P_4 obtained.

$$n_P = 10 + 8 + 2 = 20 \text{ mol}$$
$$n_{P_4} = \frac{20}{4} = 5 \text{ mol}$$

Q. Find pressure exerted by helium gas (4g) taken in 10 lit vessel at 600k temperature. (Find pressure in Pa)

Sol

$$PV = nRT$$
$$P \times 10 = \frac{4}{4} \times \frac{1}{12} \times 600$$
$$P = 5 \text{ atm} = 5 \times 1.013 \times 10^5 \text{ Pa}$$

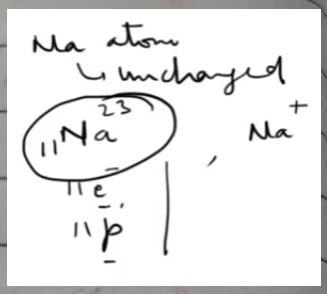
Sol?
1st approach: $PV = nRT$, $P \times 10 = \frac{4}{4} \times \frac{1}{12} \times 600$, $P = 5$ atm $= 5 \times 1.013 \times 10^5$ Pa ✓
2nd approach: $PV = nRT$, $P \times (10 \times 10^{-3}\ m^3) = 1 \times \frac{25}{3} \times 600$, $P = 25 \times 200 \times 100$ Pa $= 50 \times 10^4$ Pa $= 5 \times 10^5$ Pa ✓

Q 2.3 g of Na^+ sample.

Find. (i) no. of e^-

(ii) no. of p (iii) charge in coulombs.

$$n_{Na^+} = \frac{2.3}{23} = 0.1 \text{ mol}$$



(i) $10 \times 0.1 \times N_A = N_A$ 'e'

(ii) $11 \times 0.1 \times N_A = 1.1\ N_A$ (p)

(iii) $= +1e \times N_A \times 0.1 = 0.1\ N_A e$

$= 0.1 \times 6.022 \times 10^{23} \times 1.6 \times 10^{-19} C$

$= 0.1 \times 96500$

$= 9650\ C$

(*) Average atomic mass or average isotopic mass

$_{17}Cl^{35}$ $_{17}Cl^{37}$ $\longrightarrow$ $M_{avg} = 35.5$ amu g/mol

$$M_{avg} = \frac{\text{Total mass of all the atoms in the sample}}{\text{Total no. of atoms}} \quad (\text{amu})$$

$$M_{avg} = \frac{\text{Total mass in (g)}}{\text{Total no. of moles}} \quad (\text{g/mol})$$

Isotope: having same atomic no but diffent mass no of an elment
$_1H^1$ (Protium), $_1H^2$ (Deuterium), $_1H^3$ (Tritium) → Radioactive, C^{12}, C^{14} Radioactive

(a) % by mol abundance of isotopes is given.

e.g A + B

$$M_{avg} = \frac{x_1 m_1 + x_2 m_2 + x_3 m_3 + \ldots}{100}$$

% by mol abundance of isotopes is given,
e.g A + B
M_A M_B
x% (100−x)% by mol or by number
$M_{av} = \frac{x \times M_A + (100-x) \times M_B}{100}$, $M_{av} = \frac{x_1 M_1 + x_2 M_2 + x_3 M_3 + \ldots}{100}$

NOTE : In a mixture % by mol of a component need not be same as % by mass of that component

eg. A + B
↓ ↓
x% (100-x)% by mol

$$\% \text{ by mass of } A = \frac{\text{mass of } A}{\text{Total mass}} \times 100$$

$$= \frac{x \times M_A}{x \times M_A + (100-x) \times M_B} \times 100$$

$$= x \times \frac{M_A}{\frac{x M_A + (100-x) \times M_B}{100}}$$

$$= x \times \frac{M_A}{M_{avg}}$$

∴ % by mass of component(A) = % by mol(A) × $\frac{M_A}{M_{avg}}$

Q: Dry air ($O_2 + N_2$)
20% 80% by mol

(i) $M_{avg} = \frac{(20 \times 32) + (80 \times 28)}{100} = 28.8$ g/mol

(ii) % by mass of $O_2 = 20 \times \frac{32}{28.8} = 22.22\%$

(b) When molar ratio of isotopes is given.

A + B
m : n molar ration

e.g. A + B
75% 25% by mole
3 : 1 molar ratio

$$M_{avg} = \frac{m \times M_A + n \times M_B}{m+n}$$

Q. A vessel contain equal masses of H_2, He and CH_4. Find M_{avg}.

H_2	He	CH_4	
$\frac{w}{2}$	$\frac{w}{4}$	$\frac{w}{16}$	mol

$$M_{avg} = \frac{w+w+w}{w/2 + w/4 + w/16} = \frac{3}{\frac{8+4+1}{16}} = \frac{48}{13} \text{ g/mol}$$

Q. A vessel has 1 mol of O_2 and 3 mol of SO_3. Find M_{avg}.

$$M_{avg} = \frac{1 \times 32 + 3 \times 80}{4} = \frac{32 + 240}{4} = \frac{272}{4} \text{ g/mol}$$

* Molecular formula :
It gives the actual number of atoms of various element in the one molecule of compound.

→ Empirical formula :
It represents the simplest whole number ratio of atoms of various elements in compound.

Eg- Glucose : $C_6H_{12}O_6$ (Molecular formula)

CH_2O (Empirical formula)
1 : 2 : 1 (Molar ratio)

→ Relation between Empirical formula & Molecular formula

$$M.F = (EF)_n$$

where $n = \frac{\text{Molecular formula weight}}{\text{Empirical formula weight}}$

E.g glucose : $C_6H_{12}O_6$ (MF)
Molecular weight = 180

Empirical formula : CH_2O
EF wt = 12 + 2 + 16 = 30

$\therefore n = \frac{180}{30} = 6$

$\Rightarrow (CH_2O)_6$ or $C_6H_{12}O_6$

→ Determination of E.F of a compound from given quantitative analysis (% by mass of various elements are known)

Step-1 Divide given masses or mass % of various elements by the atomic weight of element. This gives relative no. of moles.

Step-2 Make these relative no. of moles of various elements in simplest whole number ratio & you get E.F

Q. Organic compound : C% = 60 %
O% = 32%
H% = 8%
} by mass

Find Empirical formula

C	O	H
$\frac{60}{12} = 5$	$\frac{32}{16} = 2$	$\frac{8}{1} = 8$

5 : 2 : 1 (Molar ratio)

Empirical formula = $C_5O_2H_8$ or $C_5H_8O_2$

Q. 7.3 g of an organic compound has C, H, N, O and found to contain 3.6 g of 'C', 0.7 g of 'H', and 1.4 g of 'N'. Find Empirical formula.

C	H	N	O
3·6g	0·7g	1·4g	1·6g
12	1	14	16

0·3 : 0·7 : 0·1 : 0·1

= 3 : 7 : 1 : 1

Empirical formula = C_3H_7NO

If Molecule of compound is found to contain 3 atoms of 'N',

then MF = $(C_3H_7NO)_3$

= $C_9H_{21}N_3O_3$

or

$C_9H_{21}N_3O_3$ (MF)

Chemical Stoichiometry :

It involves calculations of amount of reactants/products involved in a chemical reaction.

In general

$aA + bB \longrightarrow cC + dD$ (Balanced)

a, b, c, d → stoichiometric coeff.

Eg. $A + 2B \longrightarrow C + D$

1 mol 2 mol 1 mol 1 mol

e.g. $Al_{1\,mol} + 3HCl_{3\,mol\,(aq)} \longrightarrow AlCl_{3(aq)} + \frac{3}{2}H_{2(g)}$

1 : 1·5

or

$2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_{2(g)}$

2 : 6 = 1 : 3 2 : 3

∵ 12 bananas cost = 50 Rs

∴ 5 bananas cost : $\frac{50}{12} \times 5$ Rs

Eg. $KClO_{3(s)} \xrightarrow{\Delta} KCl_{(s)} + 1{\cdot}5\,O_{2(g)}$ (Balanced)

5 mol $\frac{1 \times 5}{1}$ $\frac{1{\cdot}5 \times 5}{1}$ = 7·5 mol

= 5 mol KCl

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Q. Combustion of methane,
8g methane undergoes combustion in excess of O_2. Find mass of products formed. Also calculate mass of O_2 needed to do complete combustion.

Soln

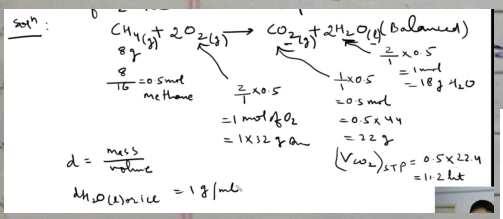
$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O \quad \text{(Balanced)}$$

$8g = 8/16 = 0.5$ mol

$\frac{2}{1} \times 0.5 = 1$ mol of O_2 $= (1 \times 32)g$

$\frac{1}{1} \times 0.5 = 0.5$ mol $= (0.5 \times 44)g = 22g$

$\frac{2}{1} \times 0.5 = 1$ mol $= 18g$ of H_2O



$d_{H_2O(l)\text{ice}} = 1$ g/ml

Volume of $H_2O(l) = \frac{\text{mass}}{d} = \frac{18}{1} = 18$ ml Ans

total

Q. Find a volume of gases evolved if 0.5 mol of lead nitrate is strongly heated to give NO_2 & O_2 gases.

$$2Pb(NO_3)_2(s) \longrightarrow 2PbO(s) + 4NO_2(g) + O_2(g) \quad \text{(Balanced)}$$

0.5 mol

$\frac{4}{2} \times 0.5 = 1$ mol

$\frac{1}{2} \times 0.5 = 0.25$ mol

Total moles $= 1.25$

$(V_{gases})_{STP} = 1.25 \times 22.4$ litres

Q. 13.5g of Al metal is dissolved in excess of H_2SO_4 Solution. Find volume of H_2 gas evolved at STP?

Soln

$$2Al(s) + 3H_2SO_4(aq) \longrightarrow Al_2(SO_4)_3(aq) + 3H_2(g)$$

$\frac{13.5}{27} = 0.5$ mol

$\frac{3}{2} \times 0.5 = 0.75$ mol

$(V_{H_2})_{STP} = (0.75 \times 22.4)$ litres

Q. Find mass of oxygen needed in kg for complete combustion of butane (1 kg)

$C_4H_{10} + 6.5O_2 \longrightarrow 4CO_2 + 5H_2O$ (Balanced)

$\frac{1000}{58}$ mol

$\frac{6.5}{1} \times \frac{1000}{58}$ mol O_2

$= \frac{6.5}{1} \times \frac{1000}{58} \times \frac{32}{1000}$ kg

$= 3.58$ kg Ans

$C_xH_yO_zN_pS_q$ ....

$+O_2$

$\longrightarrow CO_2 + H_2O + N_2\uparrow + SO_2$

✱ PROBLEMS ON MIXTURES –

→ Heating Effects : $M_2CO_3(s) \xrightarrow{\Delta}$ No decomposition

M → Na, K, Rb, Cs

Except $Li_2CO_3(s) \xrightarrow{\Delta} Li_2O(s) + CO_2\uparrow$

- $MCO_3(s) \xrightarrow{\Delta} MO(s) + CO_2\uparrow$

[M → Ca, Mg, Be, Sr, Ba, Cu]

- $M(NO_3)_2 \xrightarrow{\Delta} MO(s) + NO_2\uparrow + O_2\uparrow$

[M → Cu / Pb / Ba / Ca / etc]

→ $2NaHCO_3(s) \xrightarrow{\Delta} Na_2CO_3(s) + H_2O\uparrow + CO_2\uparrow$

$Ca(HCO_3)_2(aq) \xrightarrow{\Delta} CaCO_3\downarrow + H_2O(l) + CO_2\uparrow$

e.g. $2NaHCO_{3(s)} \xrightarrow[t>100^\circ C]{\Delta} Na_2CO_3(s) + H_2O\uparrow + CO_2\uparrow$

Imp $2NaHCO_{3(aq)} \xrightarrow{\Delta} Na_2CO_{3(aq)} + H_2O(l) + CO_2\uparrow$

$NH_4HCO_3(s) \xrightarrow{\Delta} NH_3\uparrow + H_2O\uparrow + CO_2\uparrow$

* Following bicarbonates exist in solid state :-
$NaHCO_3$, $KHCO_3$, $CsHCO_3$, $RbHCO_3$, NH_4HCO_3

Q. $Na_2CO_3 + NaHCO_3 = 100g$
↳ strongly heated

11.2 lit of CO_2 gas is obtained under STP.

Find % by mass of $NaHCO_3$ in the sample.

Soln $2NaHCO_3(s) \xrightarrow{\Delta} Na_2CO_3(s) + CO_2(g) + H_2O$
1 mol
$\frac{11.2}{22.4} = 0.5$ mol

$m_{NaHCO_3} = 1 \times 84 = 84g$

$\% NaHCO_3 = \frac{84}{100} \times 100 = 84\%$

Q $CH_4 + C_2H_4 = 5g$
↳ combustion

Max. of $\frac{44}{3}$ g of CO_2 gas is obtained.
Find % by mass of CH_4

$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

$C_2H_4 + 3O_2 \longrightarrow 2CO_2 + 2H_2O$

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Q. Sample of clay (silica + Inert impurities + Moisture)

40% 50% 10% by mass

heated ↓

Partially dried clay
(silica + impurities + Moisture)

% of moisture gets reduced to 6% due to heating.

Find % by mass of silica in dried clay.

Soln

Silica	+	Impurities	+	Moisture
x%		(94 − x)%		6% by mass

Logic : mass ratio of silica & impurities before heating & after heating must be same.

$$\frac{40}{50} = \frac{x}{94-x}$$

$$3760 - 40x = 50x$$

$$3760 = 90x$$

$$x = \frac{3760}{90}$$

$$x = 41.7\ \%$$

* Factor Label Method :

Particularly useful when sequential reaction are given.

Eg $A + 2B \longrightarrow 3C + D$

$2C + E \longrightarrow 3F + G$

$3G + Y \longrightarrow 2(Z) \rightarrow$ desired product

↑ Find amount of Z formed ?

desired product

Find mass of Z that can be obtained when 1 kg of A and excess of other reactants are taken.

Sol? $\left(\frac{1000}{M_A} \times \frac{4}{2} \times \frac{3}{2} \times \frac{3}{2}\right) \times M_Z$ = mass of Z obtained

mol of A, mol of C, mol of F, mol of Z

Ans = $\frac{1000}{M_A} \times \frac{3}{1} \times \frac{1}{2} \times \frac{2}{3} \times M_Z$ = mass of Z in g

↑ Moles of A; ↑ moles of C; ↑ moles of G; ↑ moles of Z

x kg

Q. $2NH_3 + 2.5O_2 \longrightarrow 2NO + 3H_2O$

$2NO + O_2 \longrightarrow 2NO_2$

$3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$

↑ 63 kg of HNO_3, find x

$\frac{x \times 10^3}{17} \times \frac{x}{x} \times \frac{x}{x} \times \frac{2}{3} = \frac{63 \times 10^3}{63}$ mol

$x = \frac{17 \times 3}{2} = 25.5$ kg

% yield of reaction

$$= \frac{\text{Actual amount of product obtained}}{\text{Theoretical (or stoichiometric) amount of product obtainable}} \times 100$$

Q Coal sample (w kg) containing 1 % of FeS_2 by mass on combustion, 2 mol of SO_2 gas is obtained with 40 % yield. Find w

$$2FeS_2 + 5.5O_2 \xrightarrow{40\%} Fe_2O_3 + 4SO_2 \quad \text{(Balanced)}$$

Soln

FeS_2 + Impur. — w kg

x kg

$$\frac{x \times 1000}{(56+64)} = \frac{x \times 1000}{120} \text{ mol}$$

$$\frac{4}{2} \times \frac{x \times 100}{120} \times 0.4 = 2$$

$$x = 0.3 \text{ kg}$$

$$w \times \frac{1}{1.0} = 0.3$$

$$w = 30 \text{ kg}$$

* Density (d) = $\frac{\text{mass}}{\text{volume}}$, unit SI = kg/m^3

CGS = g/cm^3 or g/ml

$d_{H_2O(l)}$ at $4^\circ C$ = 1 g/ml = 1000 kg/m^3

* Relative density = $\frac{\text{density of a substance}}{\text{density of reference substance}}$

→ Specific gravity - (used mainly for solids and liquids)

→ Vapour density - (For gases only)

- Specific gravity = $\frac{\text{density of a substance}}{\text{density of water at } 4^\circ C}$

↑ Temperature dependent

$= \frac{(d_{sub.}) \text{ in g/ml}}{1 \text{ g/ml}}$

eg. $d_{Al(s)} = 7.8 \text{ g/ml}$.

Specific gravity of Al(s) = 7.8

- Vapour density (VD) = $\frac{\text{density of gas}}{\text{density of reference gas}}$

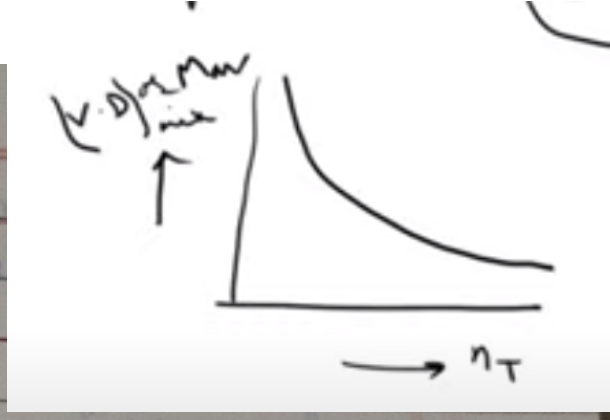
$PM = dRT$

$\Rightarrow d \propto M$

$\therefore = \frac{M_{gas}}{M_{ref\ gas}}$

Generally if not mentioned in question, ref gas is H_2

$$V_D \text{ of gas} = \frac{M_{gas}}{M_{H_2}} = \frac{M_{gas}}{2}$$



→ For a gas mixture

V.D of mixture = $\frac{M_{avg}}{2}$

Q. A vessel contains H_2, CH_4, He in mass ratio 2 : 4 : 1. Find V.D of gas mixture.

H_2	CH_4	He
2	4	1

$\frac{2}{2} : \frac{4}{16} : \frac{1}{4}$ (Mole ratio)

$1 : 1/4 : 1/4$

$4 : 1 : 1$ → mole ratio

V.D of mixture = $\frac{1}{2}\left[\frac{4\times2 + 1\times16 + 1\times4}{4+1+1}\right]$

$= \frac{1}{12} \times 28 = \frac{28}{12}$

(*) Limiting Reagent or Limiting Reactant

→ Case of L.R arise when two or more reactants are involved in a reaction.

Excess Reactant

eg. $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

1 mol — 5 mol — 2 mol

LR

5 − 1 = 4 mol

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→ Amount of product should be calculated on the basis of amount of L.R

eg. $H_2(g) + Cl_2 \longrightarrow 2HCl$
5 mol 5 mol 10 mol

In this case H_2 and Cl_2 are in stoichiometric proportion.

→ How to identify L.R

(i) First divide given no. of moles of reactants by their stoichiometric coeff. and obtain the ratio.

(ii) Reactant with least value of ratio is L.R

eg. $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$
5 mol (L.R) 5 mol
Excess reactant

N_2	H_2
$\frac{5}{1}$	$\frac{5}{3}$

Q. 10g each of Carbon and Cl_2 are taken in a vessel and made to react to form CCl_4

(i) identify L.R
(ii) Find mass of CCl_4 obtained
(iii) Find mass of excess reactant remained at the end if any.

Excess ↙ L.R ↙

Soln (i) $C + 2Cl_2 \longrightarrow CCl_4$

$\frac{10}{12}$ mol $\quad \frac{10}{71}$ mol

C	Cl_2
$\frac{10}{12\times1}$	$\frac{10}{71\times2}$

(ii) $\frac{1}{2} \times \frac{10}{71} (12 + 4\times35.5)\,g$

(iii) Mass of carbon remained

$= 20 - \frac{1}{2} \times \frac{10}{71} (12 + 4\times35.5)\,g$

OR

$W_c = \left(\frac{10}{12} - \frac{1}{2}\times\frac{10}{71}\right) \times 12\,g$

Q. 7g each of N_2 & H_2 are reacted to form NH_3

(i) Identify L.R

(ii) . . . . . (iii) . . . . .

L.R ↙ Excess ↙

Soln (i) $N_2 + 3H_2 \longrightarrow 2NH_3$

$\frac{7}{28}$ mol $\quad \frac{7}{2}$ mol

N_2	H_2
$\frac{7}{28\times1}$	$\frac{7}{2\times3}$

(ii) $\frac{2}{1} \times \frac{7}{28} \times (14+3) = 2\times0.25\times17\,g$

(iii) $(14 - 2\times0.25\times17)\,g = 5.5\,g$

OR

$W_c = \left(\frac{7}{2} - \frac{3}{1}\times0.25\right)\times2\,g = 5.5\,g$

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Grp Q. 12g 'C' reacts with 24g of oxygen to form oxides of carbon under reaction conditions. CO may form, CO_2 may form, depending upon relative amounts of C and O_2

Reaction given : $C + \frac{1}{2} O_2 \rightarrow CO$

$C + O_2 \rightarrow CO_2$

$CO + \frac{1}{2} O_2 \rightarrow CO_2$

$CO_2 + C \rightarrow 2CO$

Find composition of reaction mixture.

Soln

LR

$C + \frac{1}{2} O_2 \rightarrow CO$

$\frac{12}{12} = 1$ mol $\frac{24}{32} = \frac{3}{4}$ ↑ 1 mol

$CO = 0.25$ -0.5

$CO + \frac{1}{2} O_2 \rightarrow CO_2$

1 mol 0.25 ↑ 0.5 mol

-0.5 L.R

$= 0.5$ mol

Final composition

$C \rightarrow 0$

$O_2 \rightarrow 0$

$CO \rightarrow \frac{1}{2}$ mol

$CO_2 \rightarrow \frac{1}{2}$ mol

mixture. LR

$C + O_2 \rightarrow CO_2$

1 mol 0.75 mol ↑ 0.75 mol

↑ 0.25 mol

$CO_2 + C \rightarrow 2CO(g)$

0.75 mol 0.25 mol ↑

-0.25

$= 0.5$ mol

$\frac{2}{1} \times 0.25$

$= 0.5$ mol

finally: $C \rightarrow 0$

$O_2 \rightarrow 0$

$CO \rightarrow 0.5$

$CO_2 = 0.5$ } mol

Conclusion :

$A_2 + B_2 \rightarrow AB_x + AB_y$

Both A_2 & B_2 are completely consumed

e.g. $C + O_2 \rightarrow CO + CO_2$

O_2 consumed

e.g. $P_4 + O_2 \rightarrow P_4O_6 + P_4O_{10}$

O_2 consumed completely

* POAC (principal of Atom Conservation)

→ another form of law of conservation of mass.

→ Balancing of reaction is not needed.

→ To be applied on L.R or on an element which is sure to be completely converted into products.

→ Write all the reactants and products in single reaction

eg. $KClO_3(s) \xrightarrow{\Delta} KCl(s) + O_2(g)$

w_g ? ?

Applying POAC on Cl

no. of atom of Cl in $KClO_3$ = no. of atoms of Cl in KCl

no. of mol of Cl in $KClO_3$ = no. of moles of Cl in KCl

$$1 \times n_{KClO_3} = 1 \times n_{KCl}$$

$$\frac{1 \times w}{39+35.5+48} = 1 \times \frac{w_{KCl}}{39+35.5}$$

POAC on 'O'

no. of moles of O in $KClO_3$ = no. of moles in O_2

$$3 \times n_{KClO_3} = 2 \times n$$

$$= \frac{3 \times w}{39+35.5+48} = 2 \times \frac{w_{O_2}}{32}$$

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Q. W g pot. oxalate ($K_2C_2O_4$) is made to react with a series of reagents to form a complex compound $K_2Zn_3[Fe(CN)_6]_2$. If carbon of $K_2C_2O_4$ is conserved, then find mass of complex compound formed?

$$K_2C_2O_4 \longrightarrow K_2Zn_3[Fe(CN)_6]_2$$

POAC on 'C'

$$2 \times n_{K_2C_2O_4} = 12 \times n_{complex}$$

$$2 \times \frac{w}{m_1} = 12 \times \frac{x}{M_2}$$

$$x = \frac{wM_2}{6M_1}$$ Ans.

Q. In what molar ratio P_4 and O_2 should be taken in a reaction vessel to form P_4O_6 and P_4O_{10} in 2:3 molar ratio.

Sol.

$$P_4 + O_2 \longrightarrow P_4O_6 + P_4O_{10}$$

POAC on 'P'

$$4 \times n_{P_4} = 4 \times n_{P_4O_6} + 4 \times n_{P_4O_{10}}$$

$$n_{P_4} = 2 + 3 = 5 \text{ mol}$$

POAC on 'O'

$$2 \times n_{O_2} = 6 \times n_{P_4O_8} + 10 \times n_{P_4O_{10}}$$

$$2 \times n_{O_2} = 6 \times 2 + 10 \times 3$$

$$n_{O_2} = 42/2 = 21 \text{ mol}$$

$$\therefore \frac{n_{P_4}}{n_{O_2}} = \frac{5}{21} \quad \text{Ans}_2.$$

(*) Various Concentration terms :-

Solution (A homogeneous mixture of two or more components)

- Solute → Minor component
- Solvent → Major component.

eg. a solution of X in Y
↑ ↑
that means, Solute Solvent

eg. a solution of water in ethanol
↑ ↑
Solute Solvent

eg. a solution of ethanol in water
↑ ↑
Solute Solvent

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(1) % mass by mass (% w/w) or % by mass or % by weight

$$= \frac{\text{mass of solute}}{\text{mass of solution}} \times 100$$

eg. 20% w/w aq. NaOH solution -

20g of NaOH is present in 100g of solution

(20g ↑ solute; 100g ↑ solution)

Solvent (water) = 100 - 20 = 80g

(2) % mass by volume (% w/v)

$$= \frac{\text{mass of solute (g)}}{\text{mass of solution (ml)}} \times 100$$

eg. 8% w/v of CH_3COOH (aq)

8g of CH_3COOH is present in 100 ml of solution

~~solvent~~

* In this case, amount of solvent (water) can be calculated if density of solution is known.

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③ Strength of solution in g/L

$$= \frac{\text{mass of solute in g}}{\text{volume of solution in L}}$$

④ Molarity (M)

$$= \frac{\text{no. of moles of solute}}{\text{volume of solution in L}}$$

Unit : mol/lit. or molar

1 molar
0.01 molar — Centimolar
0.1 molar — Decimolar
0.001 molar — Millimolar

Note : To convert a w/w conc. term into a w/v conc. terms and vice versa, we need density of solution.

Q. 12 g of acetic acid is dissolved in 100 ml of water to form solution. (Density = 1.1 g/ml). Find.
(i) Molarity
(ii) % w/v
(iii) % w/w

Soln (i) Molarity $= \dfrac{(12/60)\ \text{mol}}{\dfrac{(12+100\times1)}{1.1\times1000}\ \text{lit}}$

$V_{solution} = \dfrac{(12+100\times1)}{1.1}\ ml$

$= \dfrac{0.2\times1100}{112}$ Molar

(ii) % w/v $= \dfrac{12\ g}{\dfrac{(12+100\times1)}{1.1}\ ml} \times 100$

$= \dfrac{12\times1.1\times100}{112}$ %

(iii) % w/w $= \dfrac{12\ g}{(12+100\times1)\ g} \times 100$

$= \dfrac{1200}{112}$ %

(5) Molality (m)

$= \dfrac{\text{no. of moles of solute}}{\text{mass of solvent in kg}}$

Unit = mol/kg or molal

(6) Mole Fraction (X → "chi") of solute.

$= \dfrac{\text{no. of moles of solute}}{\text{total no. of moles in solution}}$

eg. (A + B)

n_A , n_B

$$X_A = \frac{n_A}{n_A + n_B}$$

$$X_B = \frac{n_B}{n_A + n_B}$$

$$X_A + X_B = 1$$

$$X_A + X_B + X_C + \cdots = 1$$

In a given phase sum of mole fraction of all components will be equal to 1.

Q. A vessel contains equal masses of H_2, He and CH_4 Find mole fraction of each component?

Soln

H_2	He	CH_4	
$\frac{w}{2}$	$\frac{w}{4}$	$\frac{w}{16}$	: molar ratio

8 : 4 : 1 molar ratio

$$X_{H_2} = \frac{8}{8+4+1} = \frac{8}{13}$$

$$X_{He} = \frac{4}{8+4+1} = \frac{4}{13}$$

$$X_{CH_4} = \frac{1}{8+4+1} = \frac{1}{13}$$

$$X_{H_2} + X_{He} + X_{CH_4} = \frac{8}{13} + \frac{4}{13} + \frac{1}{13} = 1.$$

Q. An ~~Sol~~ aq. solution of 10% w/w NaOH is present in a bottle. Density of solution is 1.1 g/ml. Find (i) Molarity (ii) Molality (iii) % w/v

Soln (i) 10g of NaOH is present in 100g of solution.

water = 100 − 10 = 90g

$$\text{(i) Molarity} = \frac{(10/40)\ \text{mol}}{\frac{100}{1.1 \times 1000}\ \text{lit}} = \frac{1}{4} \times \frac{1100}{100} = \frac{11\ \text{mole}}{4}$$

$$\text{(ii) Molality} = \frac{(10/40)\ \text{mol}}{\left(\frac{90}{1000}\right)\ \text{kg}} = \frac{1}{4} \times \frac{1000}{90} = \frac{100}{36}\ (m)$$

$$\text{(iii) \% w/v} = \frac{10\ g}{\frac{100}{1.1}\ ml} \times 100 = 11\%$$ Ans.

Q. An aq. soln of acetic acid is rated as 0.8 Molar. Find molality of solution if density of solution is 1.2 g/ml.

Soln 0.8 mol of CH_3COOH is present in 1 Lit.

$$\text{Molality} = \frac{0.8\ \text{mol}}{\underbrace{(1000 \times 1.2)}_{\text{Mass of solution}} - \underbrace{(0.8 \times 60)}_{\text{Mass of solute}}\ \text{g}} = \frac{800}{1200 - 48}\ \text{kg} = \frac{800\ (m)}{1152}$$ or mol/kg

Q. An aq. CH_3COOH Sol^n molality = 0.8 molal. Find molarity. If [density = 1.2 g/ml]

Soln 0.8 mol of CH_3COOH is present in 1000 g of water (solvent)

$$\text{Molarity} = \frac{0.8 \text{ mol}}{\left(\frac{1000 + 0.8 \times 60}{1.2 \times 1000}\right) \text{lit}} = \frac{0.8 \times 1.2 \times 1000}{1048} \text{ M}$$

$$= \frac{960}{1048} \text{ Molar}$$

Q. An aq. solution of urea (NH_2CONH_2) has mole fraction of urea equal to 0.2. Find molality of solution.

Soln 0.2 mol urea (solute)
0.8 mol water

$$\text{Molality } (m) = \frac{0.2 \text{ mol}}{\left(\frac{0.8 \times 18}{1000}\right) \text{kg}} = \frac{200}{0.8 \times 18} \text{ mol/kg}$$

Q 50 g of $CuSO_4 \cdot 5H_2O$ [Cu → 64] dissolved ~~present~~ in 200 g of water solvent to form solution of density 1.2 g/ml

Find (i) Molarity of Cu^{2+}
(ii) Molality of Cu^{2+}

Soln

$$CuSO_4 \cdot 5H_2O \xrightarrow[H_2O(l)]{} Cu^{2+}(aq) + SO_4^{2-}(aq) + 5H_2O(l)$$

(i) Molality of Cu^{2+} $= \frac{(50/250) \text{ mol}}{\left(\frac{50 + 200}{1.2 \times 1000}\right) \text{lit}} = \frac{0.2 \times 1200}{250}$

$$= \frac{24}{25} \text{ molar}$$

(ii) Molality of Cu^{2+} = $\frac{0.2 \text{ mol}}{\left(200 + \frac{50}{250} \times 5 \times 18\right) \text{ kg}}$ / 1000

$= \frac{200}{218}$ mol/kg

* Mixing of Solutions :-

Problems on dilution of solution / mixing of solutions -

Q. Find final concentration of acetic acid when
100 ml, CH_3COOH (0.1 M) + 900 ml water
↓
Final solution

Soln $[CH_3COOH]_f$ = Total moles of acetic acid / Total ~~mol~~ volume in (L)

$= \frac{0.1 \times 100 \text{ (mmol)}}{1000 \text{ ml}} = 0.01$ molar

eg. H_2SO_4 (0.2 molar)
↓
$2H^+(aq) + SO_4^{2-}(aq)$
$[H^+] = 2 \times 0.2 = 0.4$ M
$[SO_4^{2-}] = 0.2 \times 1 = 0.2$ M

Strong electrolytes : which ionise almost completely in water.
eg. Acids : HCl, HBr, HI, HNO_3, $HClO_4$, H_2SO_4 etc
Bases : NaOH, KOH, $Ba(OH)_2$ etc
Salts : NaCl, K_2CO_3, Na_2CO_3 Baele etc.

Q. Find final concentration of each ion.

(a) 100 ml, 0.1 M $H_2SO_4(aq)$ + 400 ml, 0.2 M $HCl(aq)$

(b) 100 ml, 0.1 M $HCl(aq)$ + 300 ml, 0.1 M $NaOH(aq)$

Soln (a)

$$[H^+]_f = \frac{0.1 \times 100 \times 2 + 0.2 \times 400 \times 1}{100 + 500} M = \frac{20 + 80}{500}$$

$$= 0.2 M$$

$$[SO_4^{2-}]_f = \frac{0.1 \times 100 \times 1}{500} = \frac{1}{50} M$$

$$[Cl^-]_f = \frac{0.2 \times 400 \times 1}{500} = \frac{8}{50} M$$

Soln (b)

L.R ↙ Excess ↙

$$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

10 mmol | 30 mmol | 10 mmol

$30 - 10 = 20$ mmol

$$[Na^+]_f = \frac{20 \times 1 + 10 \times 1}{400} = \frac{3}{40} M$$

$$[OH^-]_f = \frac{20 \times 1}{400} = \frac{1}{20} M$$

$$[Cl^-]_f = \frac{10 \times 1}{400} = \frac{1}{40} M$$

$$[H^+]_f \simeq 0$$

Q 100 ml, 0.1 M $AlCl_3(aq)$ + 200 ml, 0.2 M $Al_2(SO_4)_3$ aq + 200 ml water

Find concentration of each ion in final solution.

Soln $[Al^{3+}]_f = \frac{100 \times 0.1 \times 1 + 200 \times 0.2 \times 2}{500} = \frac{90}{500} = \frac{9}{50}$ M

$[SO_4^{2-}]_f = \frac{200 \times 0.2 \times 3}{500}$ M

$[Cl^-]_f = \frac{100 \times 0.1 \times 3}{500}$ M

Q. 100 ml, 0.1 M $AgNO_3(aq)$ + 100 ml, 0.2 M $KCl(aq)$

Find concentration of each ion in final solution.

Soln

	$AgNO_3(aq)$	+ KCl	→ $AgCl\downarrow$ (white ppt.)	+ $KNO_3(aq)$
L.R	10 mmol	20 mmol		
		−10	↑	↑
		= 10 mmol	10 mmol	10 mmol

$[K^+]_f = \frac{10 \times 1 + 10 \times 1}{200} = \frac{1}{10}$ M

$[Cl^-]_f = \frac{10 \times 1}{200} = \frac{1}{20}$ M

$[NO_3^-]_f = \frac{10 \times 1}{200} = \frac{1}{20}$ M

$[Ag^+]_f \approx 0$

Note: Following salts are insoluble in water.

- AgCl, AgBr, AgI, $BaSO_4$, $CaCO_3$, PbI_2 etc.

Q. What volume of 0.2 M H_2SO_4 to be mixed in 100 ml 0.2 M (KOH)aq so that final molarity of H_2SO_4 is 0.1 M.

Soln $KOH \longrightarrow$ L.R.

$$H_2SO_{4(aq)} + 2KOH(aq) \longrightarrow K_2SO_4(aq) + 2H_2O(l)$$

$$0.1 = \frac{\left[\begin{matrix}(V \times 0.2)\,\text{mmol} \\ -10\end{matrix}\right]}{V + 100} \qquad \begin{matrix}100 \times 0.2 \\ = 20\ \text{mmol}\end{matrix}$$

$$\Rightarrow \quad 0.1V + 10 = 0.2V - 10$$

$$0.1V = 20$$

$$V = 200\ \text{ml}$$

Q → 100 ml, 20% w/w $NaOH_{(aq)}$ ($d = 1.4$ g/ml)

↓ + water (V ml)

10% w/w $NaOH_{(aq)}$, $d = 1.2$ g/ml

Find V.